

Accuracy and Efficiency of a GIS Application Using Google Earth Geolocation and LiDAR Data to Reconstruct and Simulate Ice Jam Flooding on the Mohawk River, New York

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Abstract

The city of Schenectady located in New York experienced flooding from the 14th to the 15th of January 2018 due to an ice jam on the Mohawk River. This occurred due to fluctuations in seasonal temperature that caused blockage of the river and the displacement of water. This research utilizes Geographical Information System (GIS) software along with Light Detection and Ranging (LiDAR) data, publicly announced locations showing the flood damage, and the related water level at the time of the flood. The flood simulation took place in a higher resolution LiDAR Digital Elevation Model (LiDAR-DEM) as a virtual background in Schenectady. The 3D model was manually flooded several times in different subsets of the studied area in order to calculate the level of the flood. The neighborhoods close to review exhibits the highest levels of flooding (as soon in Figures 2 and 3). The standard deviation of the measured surface elevation of the water from the ice jam flood images and the Global Mapper simulation is approximately 1.1 ft. and the average offset between those two techniques is 1.8 ft. This shows that floods can be analyzed without it being necessary to visit the site, because of these virtual simulations under high resolution, such as LiDAR provide, due to the capabilities of GIS software and widely available public images of flooding.



Figure 1: Aerial image obtained from Google Earth of the study area and the utilized referenced points of flood occurrences with water level seen in publicly available pictures from the ice jam of the 14-15th of January 2018 in Schenectady, New York.

Introduction

Schenectady is a city located in Upstate New York that is situated along the Mohawk River. The areas along the river have often experienced flooding (Figure 1). Floods have become increasingly common, and on January 14-15th there were ice jams in parts of the Mohawk River. Ice jams occur when there is a fluctuation in temperature during a short period of time due to abnormally cold temperatures followed intervals of warming. Previous work has focused on spatial and time series prediction or damage evaluation at this location (Marsellos et al., 2010, Foster et al., 2011; Tsakiri et al., 2013, Lewis et al., 2016). This study employs both GIS software and LiDAR data, and the chosen points, which are pictured above in Figure 1, to simulate the floods and compare the two methods used.

Methods

- Photographic evidence of the ice jam induced flooding on the specified dates was acquired from publicly accessible sources (Figure 2), such as internet, online newspaper, and blog-media
- The surface water level from the flooding was inferred through the previously obtained images
- Google Earth was used to geo-address these areas
- The study points from Google Earth along with a LiDAR-DEM, as a virtual background, were both imported into the GIS software, Global Mapper
- A 3D model was created to simulate the flood levels in Global Mapper (Figure 3)
- Each point was flooded manually, in order to improve the accuracy of flooding based on the photographic evidence
- The simulated flood levels from Global Mapper and the obtained imagery were compared
- The elevations from Google Earth and the LiDAR elevations of simulated flooding were extrapolated for each site

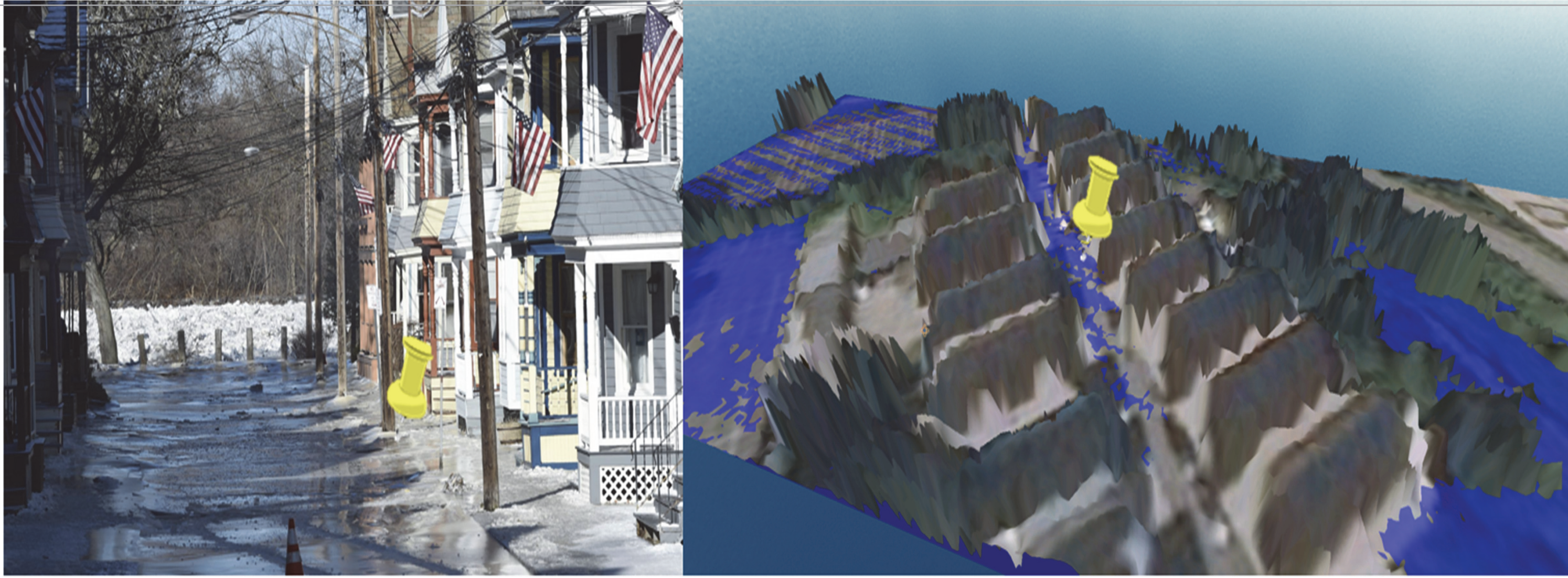


Figure 2: (left picture) Global Mapper 3D model of flooded Site 4, at Ingersoll Ave, and (right picture) the publicly available image showing the flood caused by the ice jam of January 2018 (images showing the flood were provided by Peter R. Barber for the Daily Gazette newspaper).

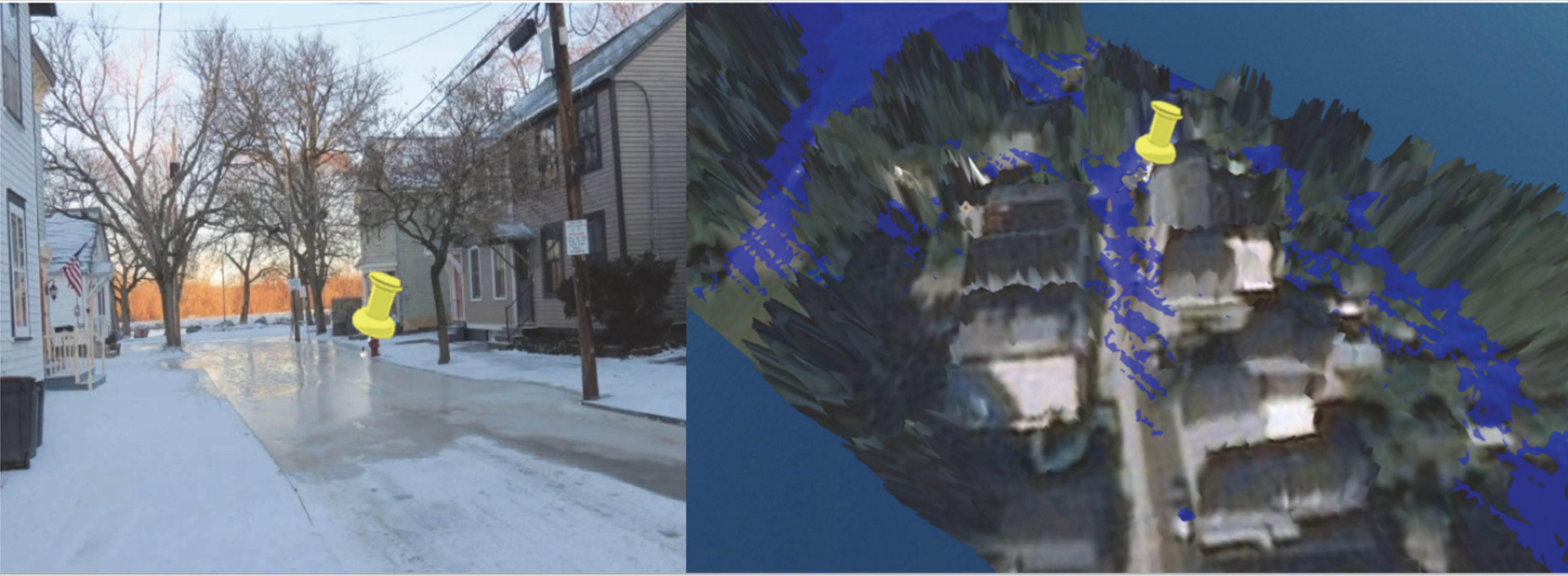


Figure 3: (left picture) Global Mapper 3D model of flooded Site 6, located at North Street, and (right picture) the publicly available image showing the flood caused by the ice jam of January 2018 (images showing the flood were provided by David Giacalone for the Suns along the Mohawk blog).

Table 1: Results and coordinates for each site.

Site	Address	Latitude	Longitude	Google Earth Elevation (feet, meters)		LiDAR (feet, meters)	
1	Jumping Jacks Dine In	42°49'19.77"	-73°57'23.53"	221.0	67.4	224.6	68.5
2	Ferry St Water Gauge	42°49'9.87"	-73°56'49.44"	221.0	67.4	223.4	68.1
3	Rivers Casino	42°49'27.89"	-73°56'9.09"	221.0	67.4	223.8	68.2
4	Ingersoll Ave: Gray and White House	42°49'13.00"	-73°56'38.61"	222.0	67.7	222.8	67.9
5	Riverside Park	42°49'16.92"	-73°56'49.70"	221.0	67.4	221.8	67.6
6	North Street: Brown House	42°49'10.83	-73°56'43.34"	223.0	68.0	223.6	68.2
7	32 Ingersoll Ave	42°49'13.99"	-73°56'40.16"	221.0	67.4	223.6	68.2
8	Riverside Park Esplanade	42°49'12.34"	-73°56'45.62"	221.0	67.4	223.0	68.0
9	Schonowee Avenue	42°49'17.24"	-73°57'5.92"	222.0	67.7	222.7	67.9

Results

- The standard deviation of the measured water elevation level was 1.1 ft and the average offset between the techniques was 1.8 ft
- LiDAR data was more accurate than Google Earth for determining elevation of studied areas and for the simulation itself
- Google Earth was more efficient and is widely accessible to the general public
- The flood damaged area in the DTM was wider than the DSM (Table 1)

Discussion

- The Bare Earth/Digital Terrain Model (DTM) was not ideal, as it did not show the accumulation of flood waters around pre-existing infrastructure
- The Digital Surface Model (DSM) was more suitable for studying the flow accumulation patterns
- Floods can be monitored and analyzed without being necessary to visit the site
- Past floods in the area can be used for reference to help predict for future floods
- This study demonstrates how similar flooding events can be studied in the future
- The improved ability to understand these implication can improve adaptive techniques

Conclusion

The comparison of LiDAR-DEM and Google Earth allowed for the creation of accurate flood simulations based on the ice jam induced flooding of Schenectady, NY (14-15th Jan 2018). This demonstrates that floods can be visualized virtually, especially using the DSM.

References

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